Scenario 4B: Low and heterogeneous exposure stress test with continuous-time x2

MSSTNB posterior performance, exposure-group recovery, and fit-status classification

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1 Purpose and main conclusion

Scenario 4B is a low and heterogeneous exposure stress test for the MSSTNB model under the same continuous-time covariate design used in Scenarios 1–3. The stress is introduced through the known exposure term rather than by directly lowering the global observation intensity. The purpose is to evaluate whether exposure-poor area-time cells contain enough information to separate regression effects, spatial effects, and latent temporal risk.

The main result is:

> Under the continuous-time x_2 design, Scenario 4B remains numerically stable and preserves accurate regression-slope recovery. The primary stress effect is not regression failure. It is weaker latent-risk recovery in low-exposure areas than in high-exposure areas.

The formal fixed-gamma sampled-λ analysis produced 9 stable fits, 1 soft-warning fit, and 0 numerical-instability fits out of 10 replicates. The one soft-warning replicate had only minimal lambda-output guard activity, and there were no numerical-instability replicates.

2 Scenario design

Scenario 4B uses the same reference dynamic latent structure as the main dynamic scenarios but modifies the known exposure. The S4B exposure is constructed as

$$e_{tj}^{\mathrm{S4B}} = e_{tj}^{\mathrm{ref}} m_{tj},$$

where m_{tj} is a low and heterogeneous exposure multiplier. The target mean multiplier is approximately 0.05, so the average exposure is approximately 5% of the reference exposure. The multipliers are heterogeneous across areas and mildly variable over time.

Table 1: Official Scenario 4B continuous-time x2 design.

Quantity	Value
Data scenario ID	S4B_LOW_HETEROGENEOUS_EXPOSURE_CONTINUOUS_X2_T100
Fit analysis	Fixed-gamma sampled-lambda fit
Stress source	Low and heterogeneous known exposure
Target mean multiplier	0.05
Area heterogeneity	area_log_sd = 0.75
Time heterogeneity	time_log_sd = 0.08
Lower multiplier bound	0.005
Upper multiplier bound	0.25
Time points	100
Coarse areas	9
x2 mode	continuous_time
x2 AR coefficient	0.50
x2 innovation SD	0.80
Fixed gamma	0.80

This design differs from S4A. S4A stresses the whole observation process by producing a global sparse-count regime. S4B stresses the model by creating local information imbalance: low-exposure areas have much weaker information than high-exposure areas even when the overall dataset is not extremely sparse.

3 Data calibration

The continuous-time S4B data-generation check passed the calibration criteria. The realized average exposure multiplier is essentially the target value, and the exposure coefficient of variation is large enough to create a clear low/high exposure contrast.

Table 2: S4B continuous-time x2 data and exposure calibration summary.

Quantity	Value
Number of replicates	10
Average count	10.306
Average median count	4.750
Average zero proportion	0.136
Average exposure ratio	0.04937
Exposure CV	0.825
Low-exposure mean count	2.469
High-exposure mean count	20.430
Low-exposure zero proportion	0.292
High-exposure zero proportion	0.025

The average count is 10.306 and the average zero proportion is 0.136. The low-exposure group has average count 2.469 and zero proportion 0.292, while the high-exposure group has average count 20.430 and zero proportion 0.025. Thus, S4B creates a strong exposure-driven information contrast.

4 Fitting strategy

The S4B fit uses the dynamic MSSTNB structure with fixed $\gamma = 0.8$. The model estimates

$$\beta_0, \beta_1, \beta_2, \phi, \tau_\phi, r, \kappa, \lambda,$$

while γ is fixed at its truth. The split component ω is disabled in this analysis. This fixed-gamma strategy isolates low-exposure stress rather than testing weak identification of γ.

Numerical guards are recorded and used to classify each replicate. The classification rule is:

- **Stable:** no beta, kappa, or lambda-output guard and no extreme latent-risk metric.
- **Soft warning:** small guard activity without severe parameter or latent-process failure.
- **Numerical instability:** large guard activity, extreme intercept, extreme lambda RMSE, or extreme log-lambda RMSE.

5 Fit-status classification

Table 3: Fit-status classification for S4B continuous-time x2.

Status	# reps	Percent
Soft warning	1	10.0
Stable	9	90.0

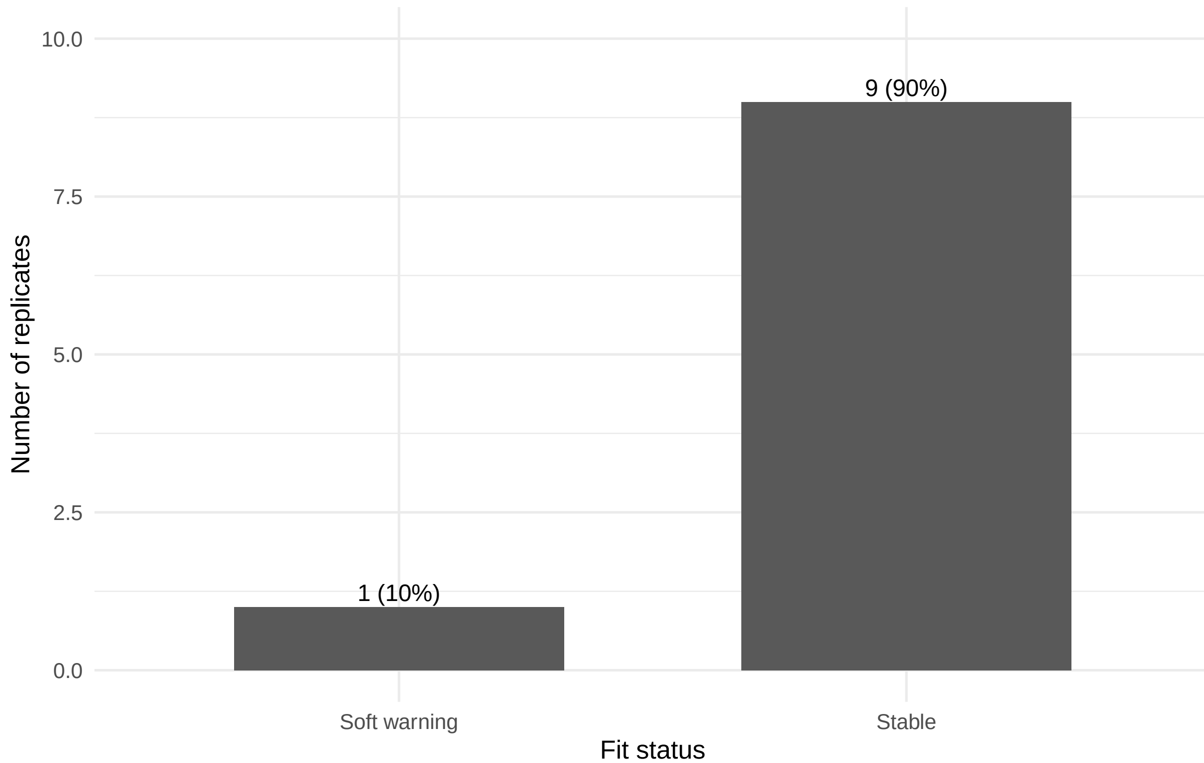

Figure 1: Fit-status counts for S4B continuous-time x2.

The fit-status result is 9 stable fits, 1 soft-warning fit, and 0 numerical-instability fits. This is an important difference from a failure-dominated stress setting: the continuous-time S4B fit is numerically stable, with only one minor warning replicate.

6 Posterior performance by subset

Because there are no numerical-instability replicates, the stable-plus-soft-warning subset is the main reporting subset. The stable-only subset is also shown as a sensitivity check.

Table 4: Data and exposure summaries by fit-status subset.

Subset	# reps	Mean count	Zero prop.	Exposure ratio	Exposure CV
All	10	10.306	0.136	0.04937	0.825

Subset	# reps	Mean count	Zero prop.	Exposure ratio	Exposure CV
Stable	9	10.730	0.137	0.04871	0.822
Stab.+soft	10	10.306	0.136	0.04937	0.825

Table 5: Regression and dispersion recovery by fit-status subset.

Subset	# reps	b1 mean	b1 cover	b2 mean	b2 SD	b2 cover	r mean
All	10	0.500	1.000	-0.400	0.015	1.000	17.585
Stable	9	0.502	1.000	-0.399	0.015	1.000	17.405
Stab.+soft	10	0.500	1.000	-0.400	0.015	1.000	17.585

The stable-plus-soft-warning posterior means are $\bar{\beta}_1 = 0.500$ and $\bar{\beta}_2 = -0.400$. The true values are $\beta_1 = 0.5$ and $\beta_2 = -0.4$. The between-replicate standard deviation of posterior means is only 0.015 for β_2, and the empirical coverage for both slopes is 1.000. Therefore, the continuous-time S4B design does not produce regression-slope fragility.

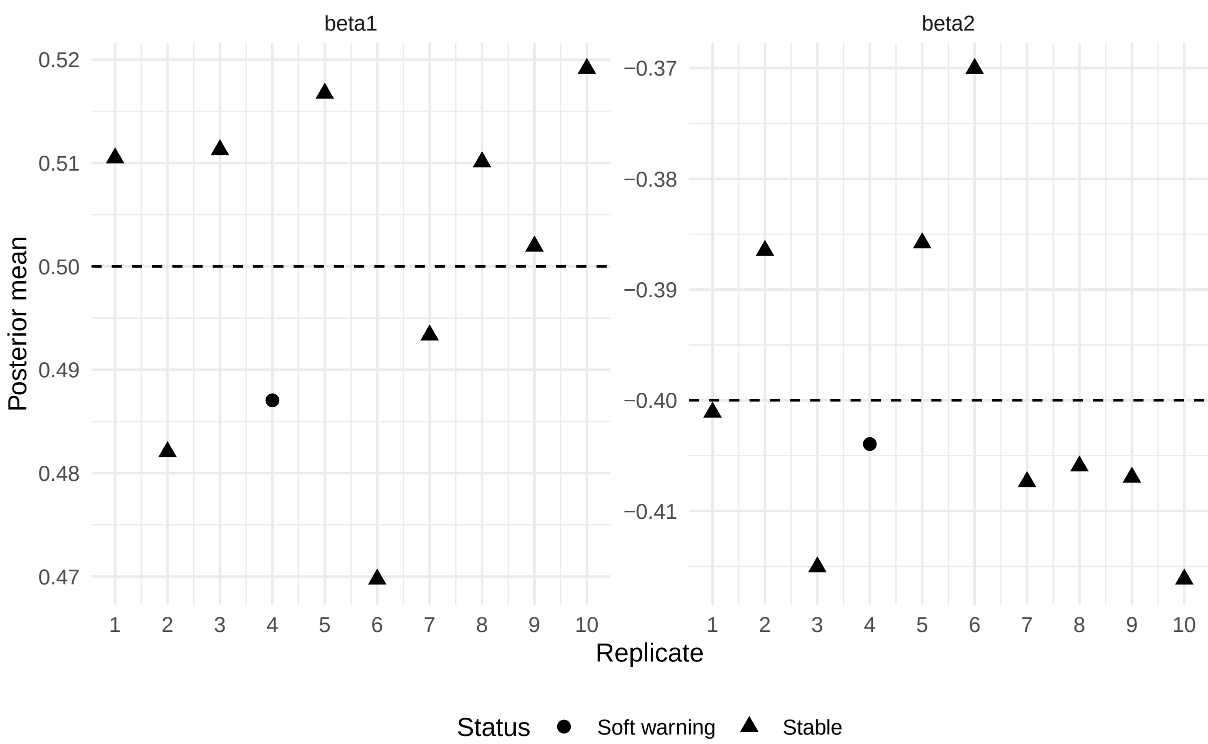

Figure 2: Posterior means of beta1 and beta2 by replicate. Dashed horizontal lines show true values.

7 Replicate-level diagnostics

Table 6: Replicate-level regression and dispersion summaries.

Rep	Status	Mean count	Zero prop.	b0	b1	b2	r
1	Stable	8.294	0.224	-2.297	0.510	-0.401	14.904
2	Stable	11.268	0.106	-1.396	0.482	-0.386	16.021
3	Stable	10.704	0.134	-1.516	0.511	-0.415	17.932
4	Soft	6.492	0.131	-2.019	0.487	-0.404	19.204
5	Stable	17.711	0.160	-1.663	0.517	-0.386	17.038
6	Stable	11.350	0.073	-1.624	0.470	-0.370	17.826
7	Stable	10.279	0.148	-1.694	0.493	-0.407	18.867
8	Stable	9.139	0.188	-1.803	0.510	-0.406	15.539
9	Stable	9.359	0.113	-1.672	0.502	-0.407	17.867
10	Stable	8.468	0.084	-1.625	0.519	-0.416	20.653

Table 7: Replicate-level latent-risk recovery and lambda-output guards.

Rep	Status	lambda RMSE	log RMSE	cor(log)	low log RMSE	high log RMSE	lambda guard
1	Stable	2.024	0.665	0.383	0.306	0.065	0
2	Stable	0.151	0.141	0.179	0.208	0.080	0
3	Stable	0.169	0.143	0.413	0.212	0.065	0
4	Soft	0.271	0.233	0.536	0.106	0.091	1
5	Stable	0.222	0.197	0.401	0.319	0.078	0
6	Stable	0.130	0.129	0.470	0.179	0.091	0
7	Stable	0.147	0.135	0.583	0.187	0.110	0
8	Stable	0.213	0.194	0.601	0.282	0.155	0
9	Stable	0.158	0.140	0.550	0.210	0.075	0
10	Stable	0.122	0.117	0.662	0.135	0.109	0

Replicate 1 has the largest global log-lambda RMSE, but it has no numerical guards and accurate slope recovery. Replicate 4 is the only soft-warning replicate, with exactly one lambda-output guard. Neither replicate should be interpreted as a full numerical failure.

8 Low- versus high-exposure recovery

The central S4B diagnostic is the contrast between low- and high-exposure areas.

Table 8: Low- versus high-exposure count summaries.

Subset	N	Low mean count	High mean count	Low zero prop.	High zero prop.
All	10	2.469	20.430	0.292	0.025
Stable	9	2.166	21.422	0.318	0.028
Stab.+soft	10	2.469	20.430	0.292	0.025

Table 9: Low- versus high-exposure latent-risk and spatial recovery contrasts.

Subset	N	Low log RMSE	High log RMSE	Low-high	Low phi	High phi
All	10	0.215	0.092	0.123	0.104	0.065
Stable	9	0.227	0.092	0.134	0.112	0.065
Stab.+soft	10	0.215	0.092	0.123	0.104	0.065

In the stable-plus-soft-warning subset, low-exposure areas have average count 2.469 and zero proportion 0.292, while high-exposure areas have average count 20.430 and zero proportion 0.025. The corresponding log-lambda RMSE is 0.215 in low-exposure areas and 0.092 in high-exposure areas.

Thus, even when the full MCMC fit is stable, local latent temporal-risk recovery is substantially weaker in low-exposure areas.

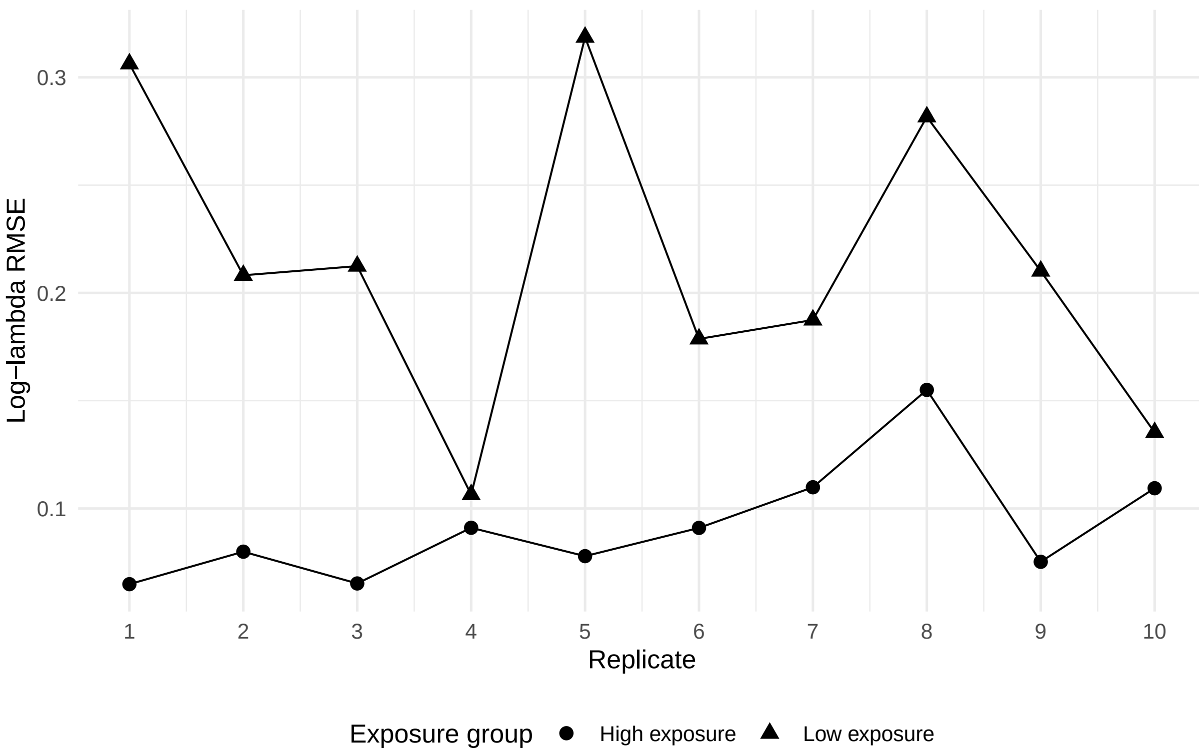

Figure 3: Low- versus high-exposure log-lambda RMSE by replicate.

9 Latent temporal-risk recovery

Table 10: Global latent temporal-risk and spatial-effect recovery by subset.

Subset	N	lambda RMSE	log RMSE	cor(log)	phi RMSE	phi cor
All	10	0.361	0.209	0.478	0.098	0.985
Stable	9	0.371	0.207	0.471	0.101	0.984
Stab.+soft	10	0.361	0.209	0.478	0.098	0.985

The stable-plus-soft-warning mean log-lambda RMSE is 0.209, and the mean correlation for log lambda is 0.478. This is clearly weaker than regression-slope recovery and weaker than the reference S3 control if available. The main S4B stress therefore appears in latent-risk recovery rather than in the regression coefficients.

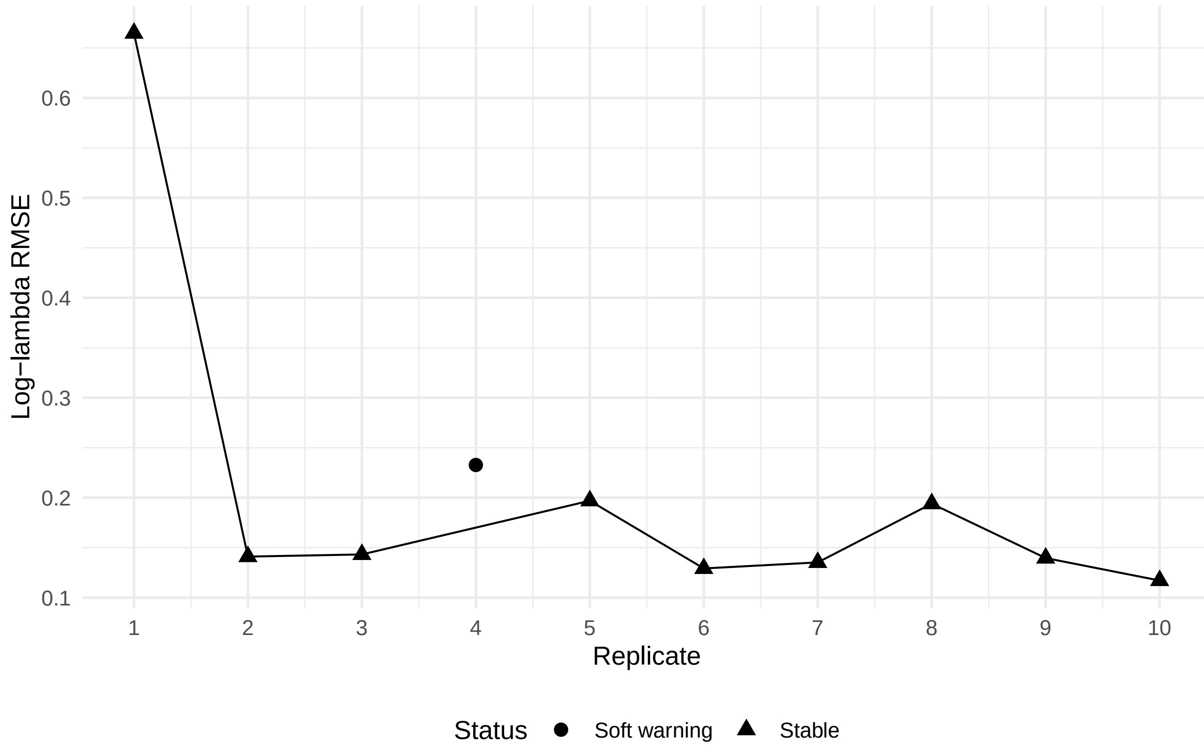

Figure 4: Global log-lambda RMSE by replicate.

10 Numerical guards

Table 11: Numerical guard summary by subset.

Subset	N	Beta guards	Kappa guards	Lambda-output guards
All	10	0	0	1
Stable	9	0	0	0
Stab.+soft	10	0	0	1

There were no beta guards and no kappa guards. The only guard activity was one lambda-output guard in the soft-warning replicate. This confirms that the continuous-time S4B result is not a numerical failure scenario.

11 Comparison with the S3 control

Table 12: Comparison of S3 control and S4B continuous-time exposure stress.

Scenario	Group	N	Mean count	Zero prop.	b1 mean	b2 mean	log RMSE	cor(log)
S3 control	All	3	195.05	0.000	0.492	-0.408	0.047	0.734
S4B exposure	Stab.+soft	10	10.31	0.136	0.500	-0.400	0.209	0.478

Relative to the S3 control, S4B preserves regression-slope recovery but has weaker latent temporal-risk recovery. This is the expected pattern for a low-exposure stress test: the exposure term is known, but low-exposure cells provide less information about local risk.

12 Interpretation of the stress mechanism

The continuous-time S4B results separate two phenomena.

First, regression slopes are recoverable. Both β_1 and β_2 remain close to their true values across the stable-plus-soft-warning subset. This is materially different from a slope-recovery failure.

Second, latent-risk recovery is exposure dependent. Low-exposure cells have smaller counts and more zeros, so they provide weaker information about the local latent temporal path. This produces higher log-lambda RMSE and higher phi RMSE in low-exposure areas than in high-exposure areas.

The posterior geometry can still be described as a local information-imbalance problem. The model does not collapse numerically under this continuous-time design, but exposure-poor areas have less information for separating

$$\beta^\top x_{tj}, \quad \phi_j, \quad \lambda_{tj}.$$

Therefore, the main S4B conclusion is not that the model fails. Rather, S4B identifies where uncertainty and weak recovery concentrate: in the low-exposure parts of the spatial-temporal domain.

13 Practical implications

The S4B continuous-time x_2 results suggest several practical implications for MSSTNB applications.

1. Exposure heterogeneity should be treated as a local information stress, even when the global count level is moderate.
2. Low- and high-exposure areas should be diagnosed separately; overall lambda summaries can hide exposure-specific weakness.
3. Regression-slope recovery can remain accurate even when latent-risk path recovery is weak.
4. Fit-status classification remains useful, but in this setting it mainly documents stability rather than failure.
5. Real applications with strongly heterogeneous exposure may need exposure-specific diagnostics, stronger priors for latent effects, or cautious interpretation of latent-risk paths in low-exposure areas.

14 Final conclusion

Scenario 4B with continuous-time x_2 confirms that low and heterogeneous exposure creates a distinct local information-imbalance stress regime for the MSSTNB model. The average exposure is approximately 5% of the reference exposure, and the exposure distribution is strongly heterogeneous. The resulting data have average count 10.31 and zero proportion 0.136.

Under this regime, 9 of 10 fitted replicates are stable, 1 replicate has a soft numerical warning, and no replicate shows numerical instability. The regression slopes are accurately recovered: the stable-plus-soft-warning posterior means are 0.500 for β_1 and -0.400 for β_2, compared with truths 0.5 and -0.4.

The main degradation occurs in latent temporal-risk recovery, especially in low-exposure areas. Low-exposure log-lambda RMSE is 0.215, while high-exposure log-lambda RMSE is 0.092. Therefore, S4B should be interpreted as an exposure-specific latent-risk recovery stress test, not as a regression-slope failure or a beta/r implementation failure.

15 Recommended reporting language

Scenario 4B evaluated the MSSTNB model under low and heterogeneous known exposure while using the same continuous-time x_2 covariate design as the main dynamic simulations. The resulting data had an average count of approximately 10.31 and an average zero proportion of approximately 0.14. In the fixed-gamma sampled-λ fit, 9 of 10 replicates were stable, 1 replicate produced a soft numerical warning, and no replicate showed numerical instability. Both regression slopes were accurately recovered, with posterior means close to the true values $\beta_1 = 0.5$ and $\beta_2 = -0.4$. The main stress effect appeared in the latent temporal-risk process: low-exposure areas had larger log-lambda RMSE than high-exposure areas. These results indicate that low and heterogeneous exposure creates a local information-imbalance regime in which regression effects remain identifiable but latent-risk recovery is weaker in exposure-poor areas.